

DIPLOMA PROJECT REPORT · GOVERNMENT GIRLS POLYTECHNIC, SHAMLI

AI-Based Smart Attendance Management System

Face-recognition attendance automation — with manual correction, supervised self-marking, leave, timetable, reporting and a full audit trail

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AGENDA

What this presentation covers

01

The Problem & Objectives

Why paper attendance fails, and what the project set out to fix

02

System Architecture & Design

Layers, data flow, use cases and the 24-table schema

03

Face Recognition Pipeline

Dual-backend recognition, confidence scoring, dataset capture

04

Three Attendance Routes

Face recognition, manual marking, supervised self-marking

05

Security, Reports & Analytics

RBAC, audit trail, 11 report types, live dashboards

06

Interface, Testing & Results

Representative screens, test outcomes, achievements

BACKGROUND

Paper attendance has three failure modes

5-10 min

lost every single class period calling out a roll-number register by hand

0%

of proxy marks are detectable after the fact in a paper register

75%

mandatory attendance threshold — students typically find out they're below it only near the exam-eligibility deadline

Government Girls Polytechnic, Shamli — three diploma branches, paper registers per subject per section, consolidated by hand every month.

OBJECTIVES

Six objectives drove every design decision

1

Automate attendance with face recognition on hardware colleges already own — a webcam, nothing more

2

Make proxy attendance structurally difficult, not just discouraged

3

Keep a manual correction path that never fights the automatic one

4

Let students self-mark, but never let their own click count without teacher approval

5

Model the whole academic structure as data — no hard-coded branches or semesters

6

Deliver institution-grade security, reporting and a tamper-evident audit trail

Why face recognition, and not the alternatives

Method	Extra hardware	Proxy-resistant	Offline
Paper register	None	No	Yes
RFID card	Reader + cards	No — can be lent	Yes
Barcode / QR	Printer / scanner	No — can be shared	Yes
Fingerprint biometric	Per-room scanner	Partially	Yes
Mobile OTP self-check-in	Student smartphones	No — self-reported	No
Face recognition (this project)	Webcam (built-in)	Yes	Yes

Three complementary routes, one attendance record

Route 1

Face Recognition

Primary route. Camera recognises enrolled students continuously and marks Present/Late automatically with a confidence score.

Route 2

Manual Marking

One-click P/A/L/LV/ML by the teacher. Overriding a face-mark requires a written, audited reason.

Route 3

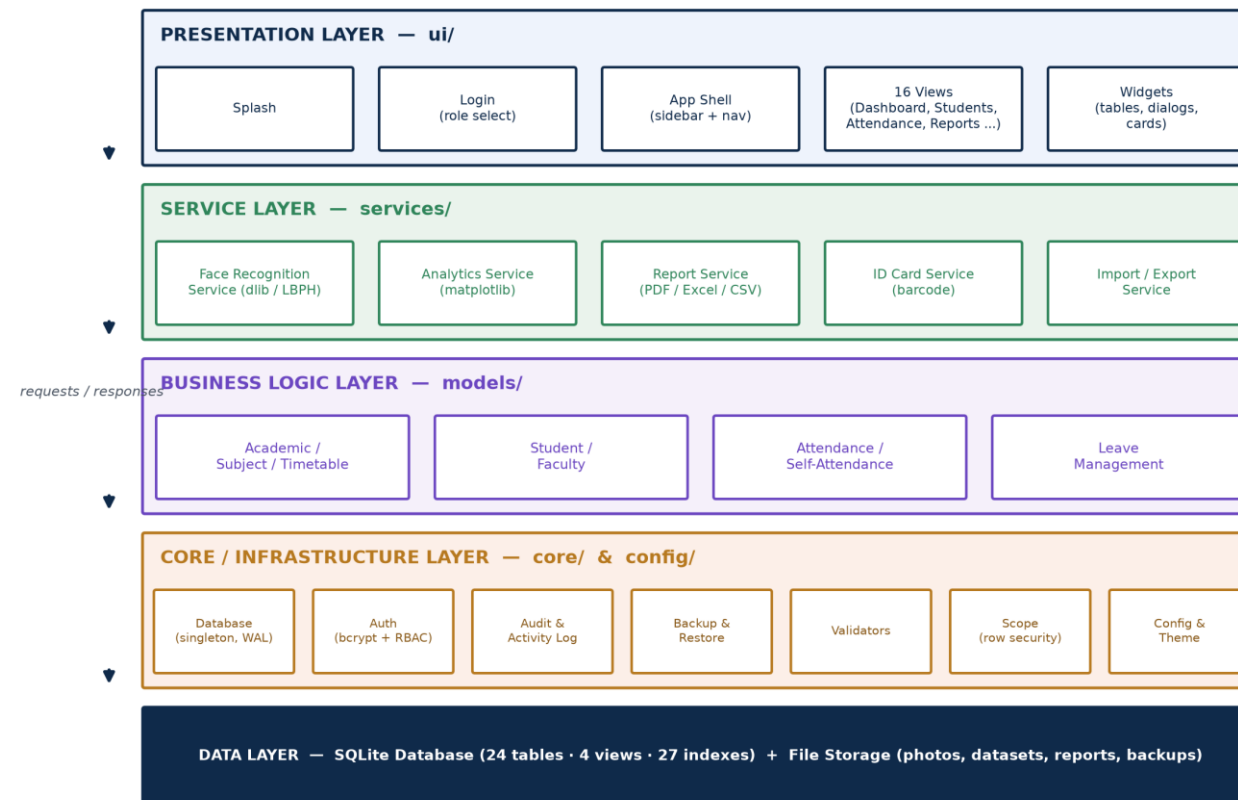
Self-Marking + Verification

A student's tap only raises a Pending request — a faculty approval is what actually changes the record.

All three write to the same attendance table — a `UNIQUE(session, student)` database constraint is what actually prevents double-marking.

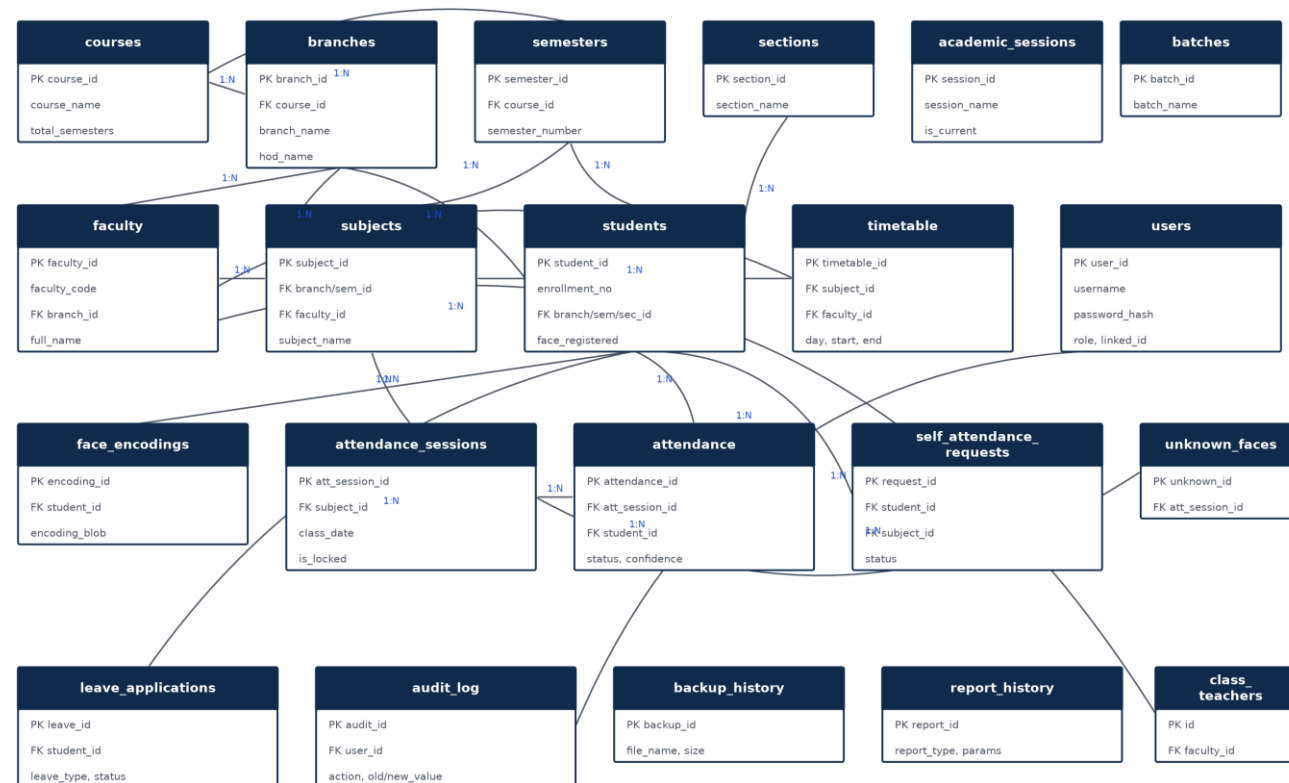
A strictly layered architecture

Fig. 4.1 Layered System Architecture



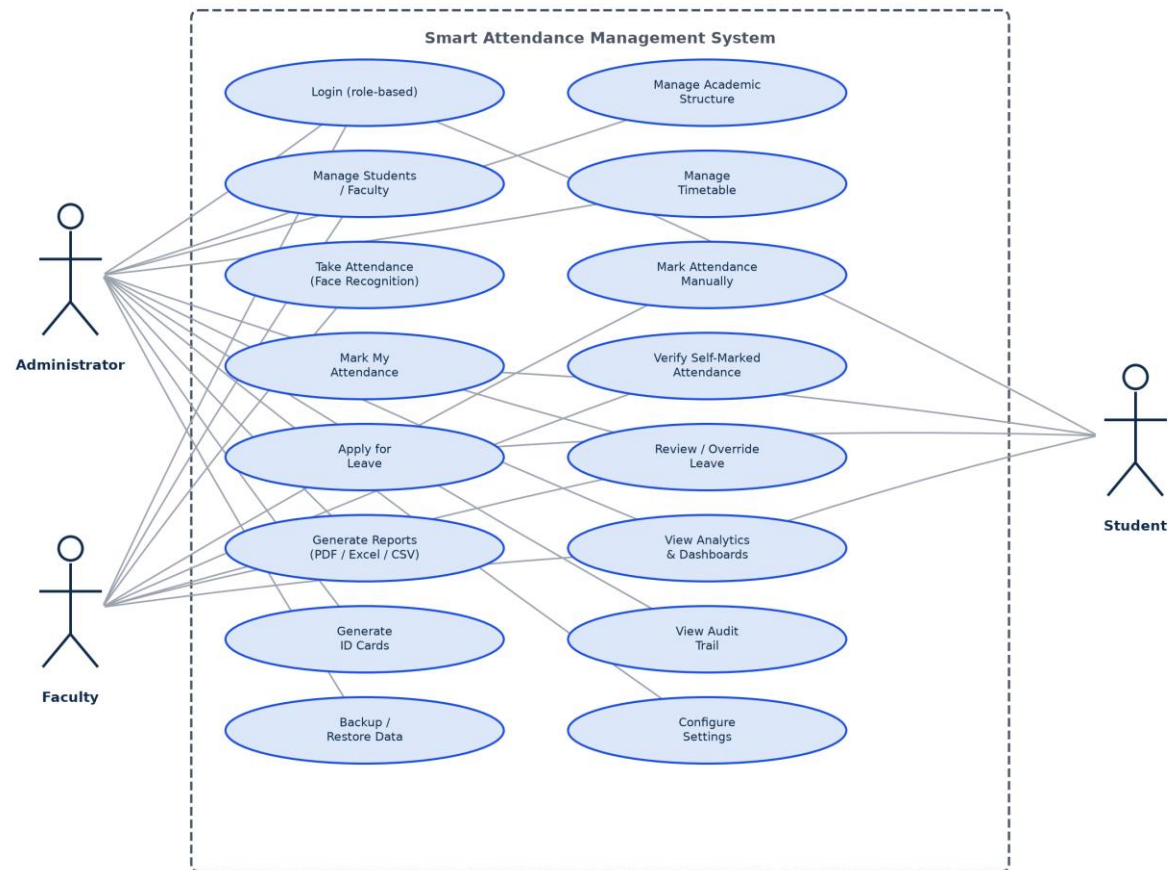
24 tables · 4 views · 27 indexes

Fig. 4.4 Entity-Relationship Diagram (core entities)



Sixteen use cases across three roles

Fig. 4.5 Use Case Diagram



Built entirely on free, open-source tooling



Interface

CustomTkinter, tkcalendar



Data & Imaging

NumPy, pandas, Pillow



Storage

SQLite — WAL mode, native backup API



Face Recognition

OpenCV LBPH · dlib / face_recognition (optional)



Reporting

ReportLab (PDF), OpenPyXL (Excel), Matplotlib (charts)



Security

bcrypt (cost factor 12)

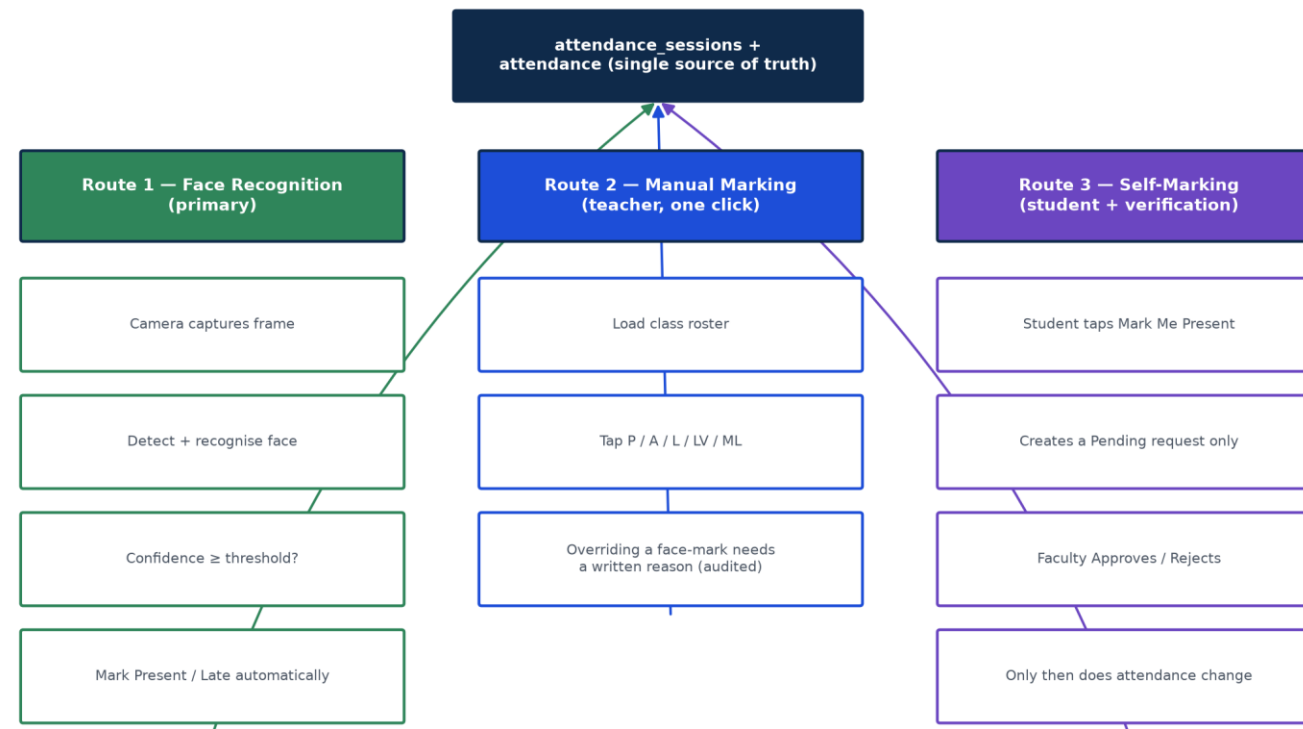
The face recognition pipeline, end to end



Dual-backend design: dlib/face_recognition (128-d embeddings, highest accuracy) is used automatically when installed; the system falls back transparently to OpenCV LBPH — no C++ compiler needed — on any ordinary college PC.

One record, three ways to reach it

Fig. 6.3 Three Complementary Attendance Routes



All three routes converge on the same attendance record — one physical row per (session, student), enforced by a UNIQUE database constraint.

Defence in depth: hashing, RBAC, scope, audit

Permission	Admin	Faculty	Student
Take attendance (face recognition)	✓	✓	—
Unlock a locked attendance session	✓	—	—
Override a leave decision	✓	—	—
Verify self-attendance requests	✓	✓	—
Mark own attendance (self-request)	—	—	✓

Below the navigation layer: core.scope restricts every query to a faculty member's own subjects/classes and a student's own records — enforced in the data layer, not just hidden in the UI.

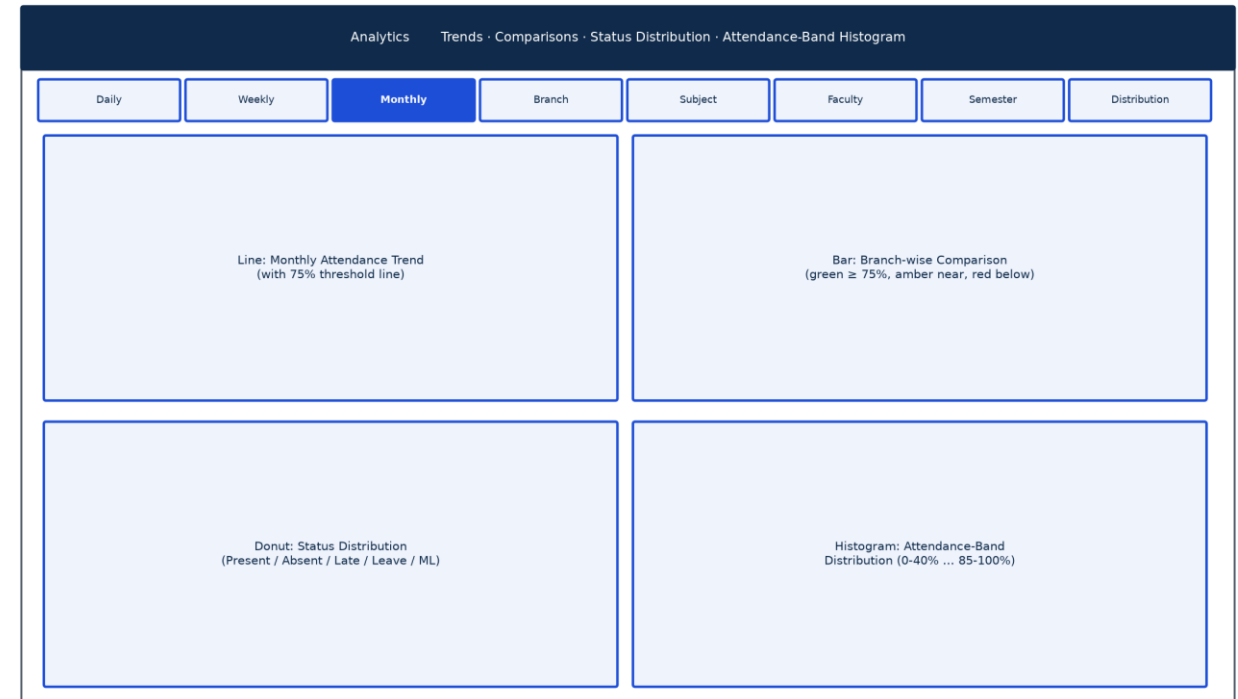
Every sensitive action is audited: attendance edits, leave overrides, lock/unlock, backup/restore — with before/after values and a mandatory reason where applicable.

Eleven report types, four chart families

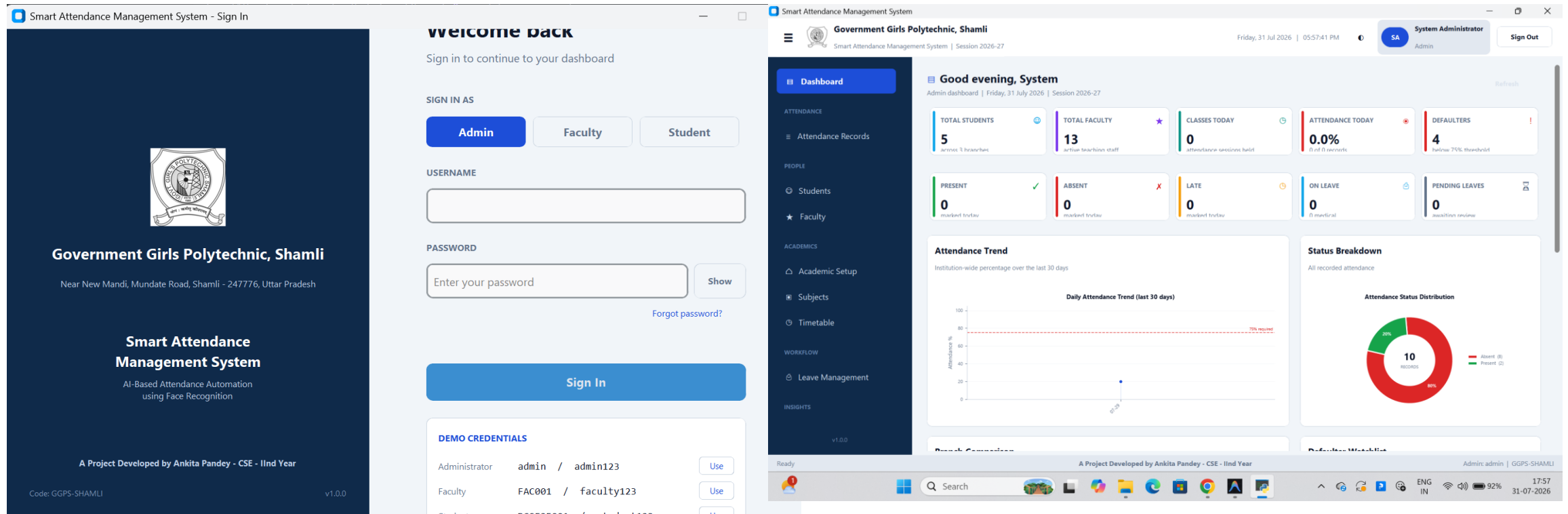
Every export in PDF, Excel or CSV — with

college letterhead, applied filters, and the generating user's name.

- Daily / Weekly / Monthly / Semester Attendance
- Student / Subject / Branch / Faculty Attendance
- Defaulter List (below eligibility threshold)
- Leave Report · Medical Leave Report
- Printable Attendance Sheet (signature register)



Sign-in and the Administrator dashboard



Role selection is a real access control — the wrong role for a valid account is rejected.

Every role sees what matters to them

The image displays two side-by-side screenshots of the Smart Attendance Management System interface, showing the dashboard for a Faculty member and a Student.

Faculty Dashboard (Left):

- Header:** Government Girls Polytechnic, Shamli. Date: Friday, 31 Jul 2026. Time: 06:00:16 PM. User: Mr. Sanjay Kumar (Faculty).
- Dashboard:** Faculty dashboard | Friday, 31 July 2026 | Session 2026-27.
- Statistics:**
 - TODAY'S CLASSES: 2 (0 still unmarked)
 - COMPLETED: 0 (attendance recorded)
 - PENDING: 2 (attendance not yet taken)
 - TO VERIFY: 1 (classific marked themselves)
 - LEAVE REQUESTS: 0 (available user online)
 - MY SUBJECTS: 2 (11 weeks remaining)
 - STUDENTS TAUGHT: 5 (attend your classes)
 - CLASSES CONDUCTED: 0 (all time)
 - TAKE ATTENDANCE: Go (marks the session)
 - MARK MANUALLY: Go (manually Present)
- Today's Schedule:** Friday, 31 July 2026.
 - 11:20 - 12:10: Data Structures Using C (CS301) (CSE - Semester 3 Sec None | Room CR-101) [Missed]
- Left Sidebar:** Dashboard, ATTENDANCE (Take Attendance, Manual Attendance, Verify Attendance, Attendance Records), PEOPLE (Students), ACADEMICS (Timetable), WORKFLOW (Leave Management), INSIGHTS.
- Footer:** Ready. A Project Developed by Ankita Pandey - CSE - IInd Year. Faculty: FAC001 | GGPS. Ready.

Student Dashboard (Right):

- Header:** Government Girls Polytechnic, Shamli. Date: Friday, 31 Jul 2026. Time: 06:02:25 PM. User: Ankita Pandey (Student).
- Dashboard:** Student dashboard | Friday, 31 July 2026 | Session 2026-27.
- Statistics:**
 - OVERALL ATTENDANCE: 75.0% (1 of 4 classes)
 - TODAY: 0/1 (1 class attended today)
 - EXAM ELIGIBILITY: Eligible (threshold 75%)
 - ABSENCES: 1 (0 late arrivals)
 - LEAVE APPLICATIONS: 1 (0 pending)
 - MARK MY ATTENDANCE: 6 (checked) you can mark now
 - AWAITING APPROVAL: 1 (sent to your teacher)
 - CLASSES TODAY: 6 (on your timetable)
 - APPLY FOR LEAVE: Go (reason: medical and more)
 - MY REPORTS: Go (attendance statements)
- My Profile:** Enrollment details on record.
 - Enrollment No: DCSE25001
 - Roll No: 1
 - Branch: Computer Science & Engineering
 - Semester: Semester 3
 - Section: -
 - Batch: 2025-2028
 - Academic Session: 2026-27
 - Face Registered: Not yet captured
- Attendance Standing:** College requirement: 75%.
 - Overall Attendance: 75.0% (3 attended | 1 absent | 0 late | 0 on leave)
 - Green bar chart showing 75% attendance.
 - Message: You are eligible to sit the examination. (Current: 75.0% | Required: 75%)
- Left Sidebar:** Dashboard, ATTENDANCE (Mark My Attendance, Attendance Records), ACADEMICS (Timetable), WORKFLOW (Leave Management), INSIGHTS (Reports), SYSTEM (Settings, About).
- Footer:** Ready. A Project Developed by Ankita Pandey - CSE - IInd Year. Student: DCSE25001 | GGPS-SHAMLI.

Faculty: today's classes and pending approvals. Student: attendance % against the 75% threshold.

Marking attendance: automatic and manual

The image displays two side-by-side screenshots of the Smart Attendance Management System interface, showing both automatic face recognition and manual attendance marking options.

Left Screenshot: Face Recognition Attendance

- Header:** Government Girls Polytechnic, Shamli. Date: Friday, 31 Jul 2026. Time: 06:00:25 PM. User: Mr. Sanjay Kumar (Faculty).
- Navigation:** Dashboard, Attendance (Take Attendance, Manual Attendance, Verify Attendance, Attendance Records), People (Students), Academics (Timetable), Workflow (Leave Management), Insights (Reports).
- Face Recognition Attendance Section:**
 - Branch: Computer Science & Engineering, Semester: Semester 3, Section: -, Subject: Data Structures Lab (CS304). **Start Attendance** button.
 - Camera status: Camera Off. Engine: OpenCV LBPH (contrib). Timer: 00:00.
 - Class Details:
 - Subject: Data Structures Lab (CS304)
 - Faculty: Mr. Sanjay Kumar
 - Class: Computer Science & Engineering - Semester 3
 - Students: 5 on roster
 - Face ready: 0 of 5 registered
 - Date: 31 Jul 2026
 - Recognition Log: Students marked in this session. Recognised students will appear here.

Right Screenshot: Manual Attendance

- Header:** Same as the left screenshot.
- Manual Attendance Section:**
 - Branch: Computer Science & Engineering, Semester: Semester 3, Section: -, Subject: Data Structures Lab (CS304), Date: 2026-07-31. **Load Class** button.
 - Class Details: Data Structures Lab (CS304) | CSE Semester 3 Sec - | 2026-07-31 | Faculty: Mr. Sanjay Kumar. Filter: All. Buttons: All Present, All Absent.
 - Summary:
 - TOTAL: 5
 - PRESENT: 0
 - ABSENT: 5
 - LATE: 0
 - LEAVE: 0
 - MEDICAL: 0
 - Roster:
 - 1. Ankita Pandey (DCSE25001) with buttons: P, A, L, LV, ML.

Footer: v1.0.0. Attendance session opened for 5 student(s). A Project Developed by Ankita Pandey - CSE - IInd Year. Faculty: FAC001 | GGPS-SHAMLI. 18:00 31-07-2026.

Live camera recognition (left) and one-click manual marking with a full P/A/L/LV/ML roster (right).

Self-marking only counts once approved — and it's all audited

The image displays two side-by-side screenshots of the Smart Attendance Management System interface, illustrating the process of self-marking and its audit trail.

Left Screenshot: Attendance Verification

The interface shows the 'Attendance Verification' section. A notification states: "These students marked themselves present. Nothing is counted until you approve it. Approving writes the attendance record and logs your decision; rejecting leaves attendance untouched. If the class attendance session, approving opens one." Below this, there are summary cards for 'AWAITING YOUR REVIEW' (1), 'APPROVED' (0), 'REJECTED' (0), and 'TOTAL REQUESTS' (1). A table lists the pending requests:

Requested	Enrollment No.	Student	Roll	Branch	Semester	Sec	Code	Subject	Class Date	Time	Status	Marked
2026-07-29 19:32	DCSE25001	Ankit Pandey	1	CSE	Semester 3	-	CS301	Data Structures Usin	29-07-2026	13:40	Pending	-

Buttons for 'Approve Selected', 'Approve as Late', 'Reject Selected', 'View Details', 'Approve All Pending', and 'Reject All Pending' are visible.

Right Screenshot: Audit Trail

The 'Audit Trail' section shows a summary of system actions:

- TOTAL ENTRIES: 31
- TODAY: 1
- ATTENDANCE EDITS: 3
- SIGN-INS: 9
- FAILED SIGN-INS: 0

Below the summary, there is a table of audit logs. The relevant entry is:

Timestamp	User	Role	Action	Module	Entity	ID	Reason
2026-07-29 19:32	DCSE25001	Ankit Pandey	ATTENDANCE_EDIT	Self Attendance	attendance_se	1	Self-attendance request #4 approved. Bulk approved by Sachin Vikal

Ankita Pandey's self-marked request awaiting approval (left); the same action logged permanently (right).

Managing students and the weekly timetable

Smart Attendance Management System

Government Girls Polytechnic, Shamli

Friday, 31 Jul 2026 | 06:01:29 PM

MS Mr. Sanjay Kumar Faculty

Sign Out

Smart Attendance Management System | Session 2026-27

Student Management

Add, edit and search student records; capture face datasets

+ Add Student Import Export Promote

TOTAL STUDENTS: 5 ACTIVE: 5 FACE REGISTERED: 0 FACE PENDING: 5

Branch: All Semester: All Section: All Batch: All Status: All Face Data: All Search:

Roll	Enrollment No	Student Name	Branch	Semester	Sec	Gender	Mobile	Batch	Face	Status
1	DCSE25001	Ankita Pandey	CSE	Semester 3	-	Female	9719812345	2025-2028	No	Active
2	DCSE25002	Priya Chauhan	CSE	Semester 3	-	Female	9719823456	2025-2028	No	Active
3	DCSE25003	Neha Salni	CSE	Semester 3	-	Female	9719834567	2025-2028	No	Active
4	DCSE25004	Shivani Tyagi	CSE	Semester 3	-	Female	9719845678	2025-2028	No	Active
5	DCSE25005	Muskan Ansari	CSE	Semester 3	-	Female	9719856789	2025-2028	No	Active

Showing 1-5 of 5 records

Student Details

Select a student from the list

AP Ankita Pandey DCSE25001

View Full Record Edit

Capture Face Dataset Clear Face Data

Generate ID Card Delete Student

Timetable

Weekly class schedule with clash detection across class, faculty and room

+ Add Slot Copy Day

Branch: Computer Science & Engineering Semester: Semester 3 Section: All Faculty: All

34 slot(s) | 28.3 hours per week

Weekly Grid Slot List

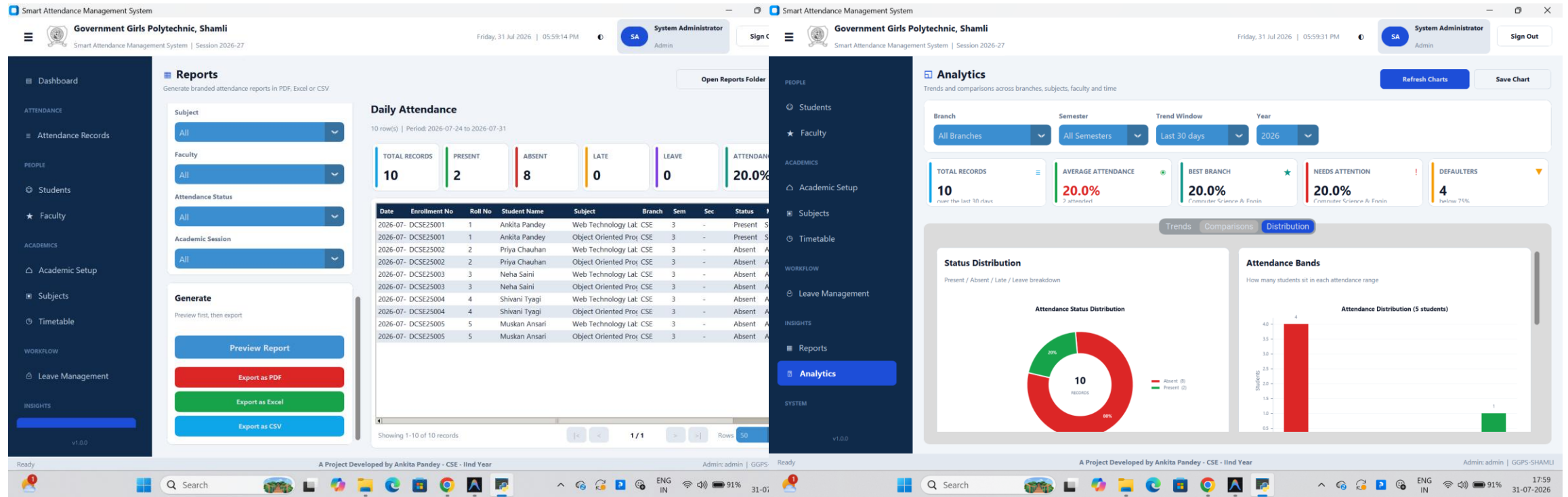
TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
09:30 10:20	CS301 Data Structures Using Mr. Sanjay Kumar Room CR-101	CS302 Object Oriented Progra Sachin Vikal Room CR-101	CS303 Digital Electronics Rashmi Room CR-101	CS304 Data Structures Lab Mr. Sanjay Kumar Room CR-101	CS305 Applied Mathematics II Swati Chauhan Room CR-101	CS306 Web Technology Lab Sachin Vikal Room CR-101
10:20 11:10	CS302 Object Oriented Progra Sachin Vikal Room CR-101	CS303 Digital Electronics Rashmi Room CR-101	CS304 Data Structures Lab Mr. Sanjay Kumar Room CR-101	CS305 Applied Mathematics II Swati Chauhan Room CR-101	CS306 Web Technology Lab Sachin Vikal Room CR-101	CS301 Data Structures Using Mr. Sanjay Kumar Room CR-101
11:20 12:10	CS303 Digital Electronics Rashmi Room CR-101	CS304 Data Structures Lab Mr. Sanjay Kumar Room CR-101	CS305 Applied Mathematics II Swati Chauhan Room CR-101	CS306 Web Technology Lab Sachin Vikal Room CR-101	CS301 Data Structures Using Mr. Sanjay Kumar Room CR-101	CS302 Object Oriented Progra Sachin Vikal Room CR-101

A Project Developed by Ankita Pandey - CSE - IInd Year

Admin: admin | GGPS-SHAMLI

5 seeded CSE students (left) and the real 34-slot, 28.3 hour/week CSE Semester 3 schedule (right).

Reports and analytics, generated live



The same 10-record window: a Daily Attendance report (left) and its Distribution chart (right).

16 functional test cases, 3 automated suites — all passing

16 / 16

representative test cases passed

3 / 3

automated smoke-test files passed (schema, imports, validators)

0

known open defects in the tested build

Representative cases

- Wrong role selected for a valid password → login rejected
- Face confidence below the minimum threshold → student not marked, logged as unknown
- Two timetable slots clash for one faculty member → save rejected before commit
- Five failed logins in a row → account locked for the configured period
- Manual override of a face-marked Present → blocked until a reason is given
- Medical leave beyond 3 days without a certificate → submission blocked

RESULTS

Every stated objective, achieved

Objective	Achieved as
Face recognition on ordinary hardware	Dual-backend recognizer, auto-selected at start-up
Proxy attendance structurally difficult	Live camera + DB uniqueness constraint on (session, student)
Manual correction that never fights automation	One-click marking; overriding a face-mark requires an audited reason
Self-marking never counts on its own	Pending request until a faculty approval mutates the record
Institution-grade reporting	11 report types + signature sheet, PDF / Excel / CSV

Limitations



Test coverage above the validator layer is manual, not an automated UI harness



No liveness detection yet — a good printed photo could, in principle, fool the camera



Single-institution, offline by design — no multi-campus sync today



dlib backend needs a C++ toolchain most classroom PCs won't have on Python 3.13+

Future scope

M

Mobile companion app for students to view attendance and apply for leave

C

Optional cloud sync across branch campuses, alongside the offline mode

L

Liveness detection / anti-spoofing to close the printed-photo gap

N

SMS / email alerts when attendance crosses below the eligibility threshold

T

Automatic, timetable-driven session opening

Q

An automated UI-level test harness

CONCLUSION

A system that succeeds by giving each manual step a narrower, auditable role — not by removing them.

23,000 lines of code · 56 modules · 24-table schema · dual-backend face recognition · role-based security with a tamper-evident audit trail — built to the standard a real institution, not just a classroom assignment, would require.

Thank You

Questions & Discussion